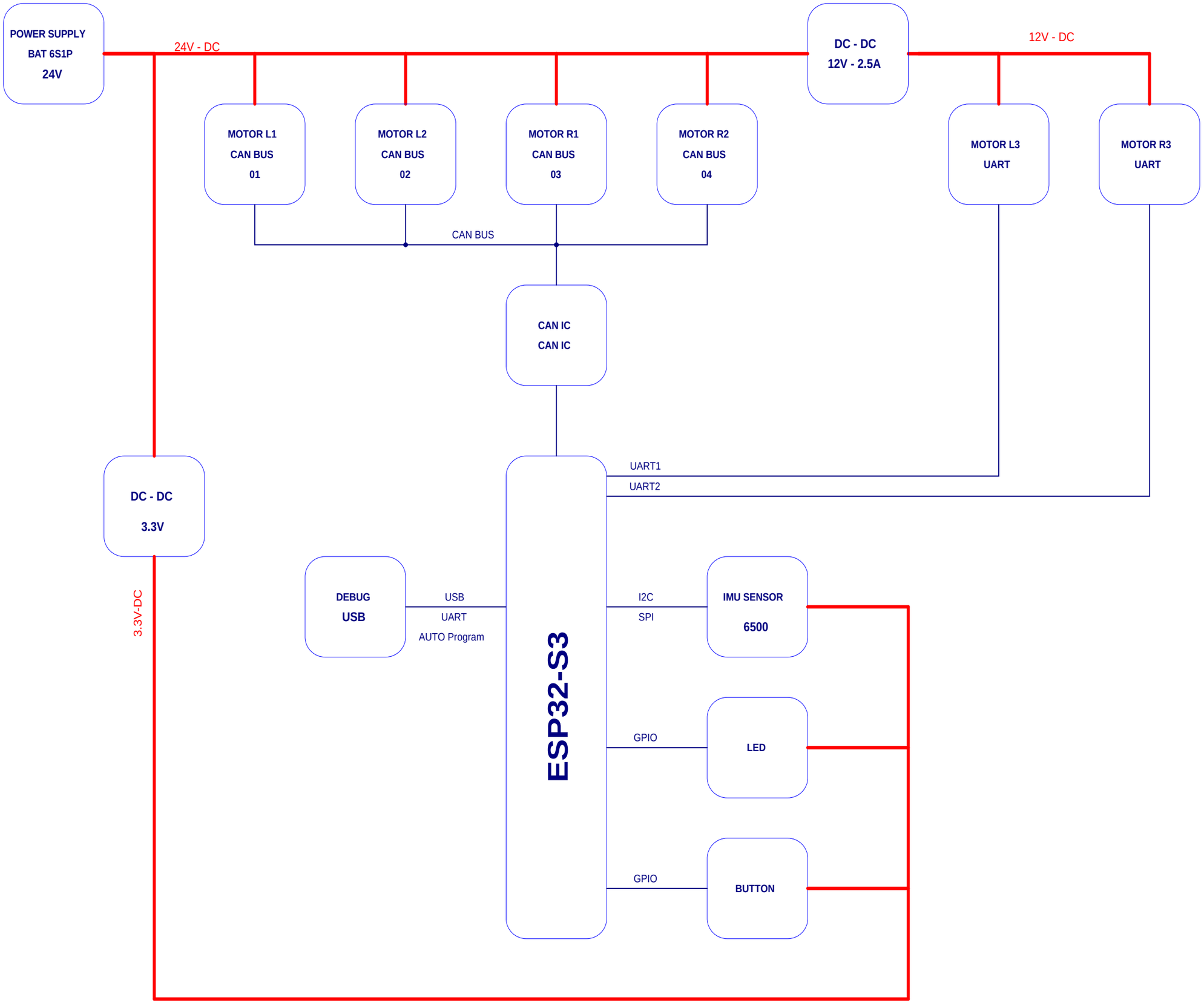
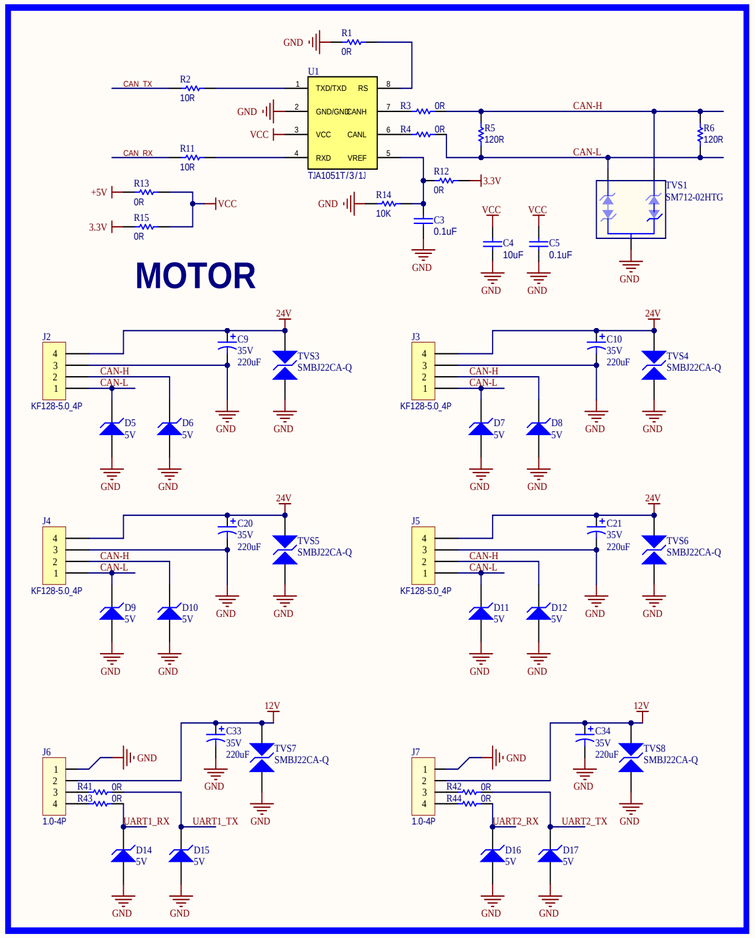
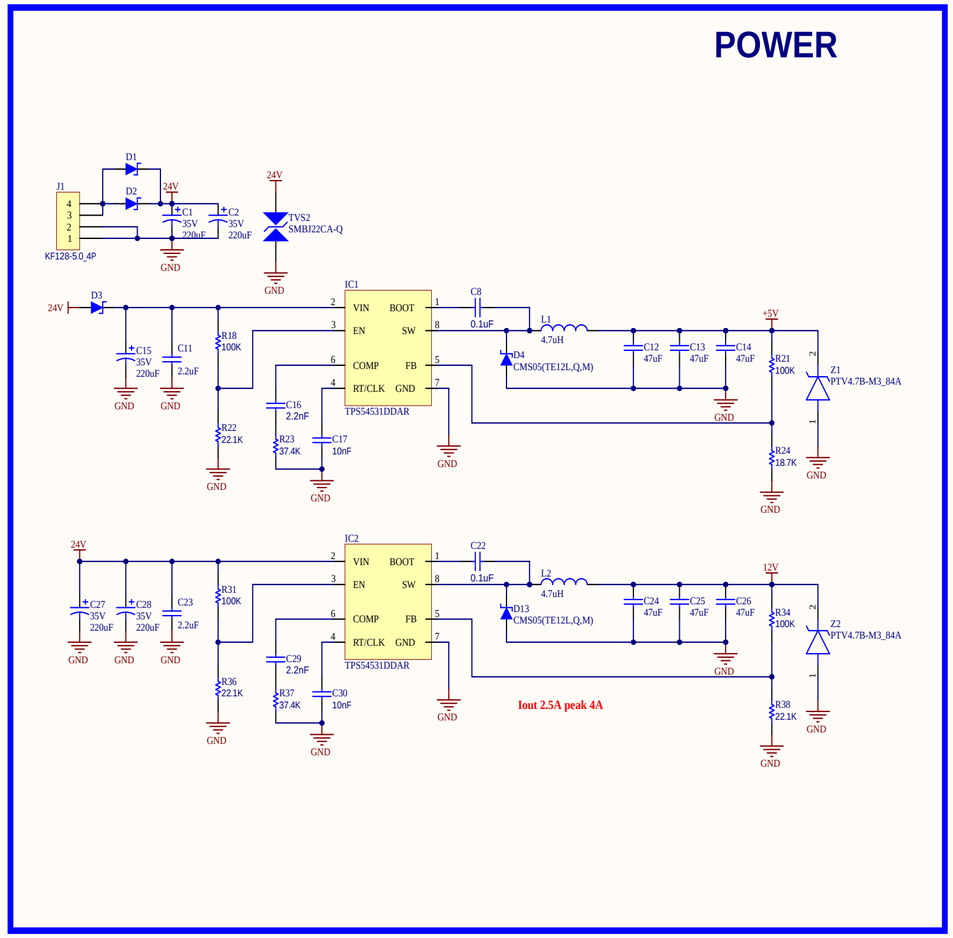
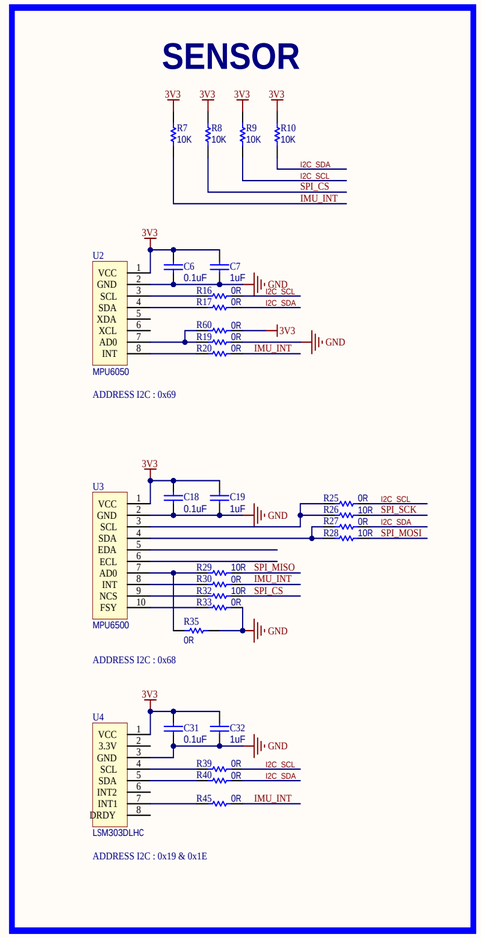
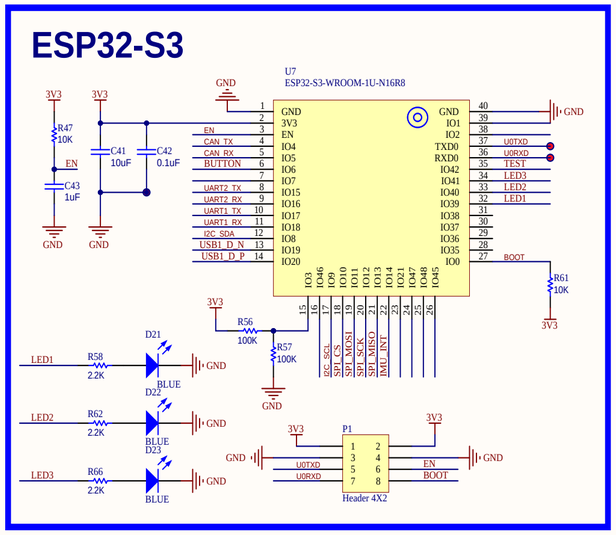
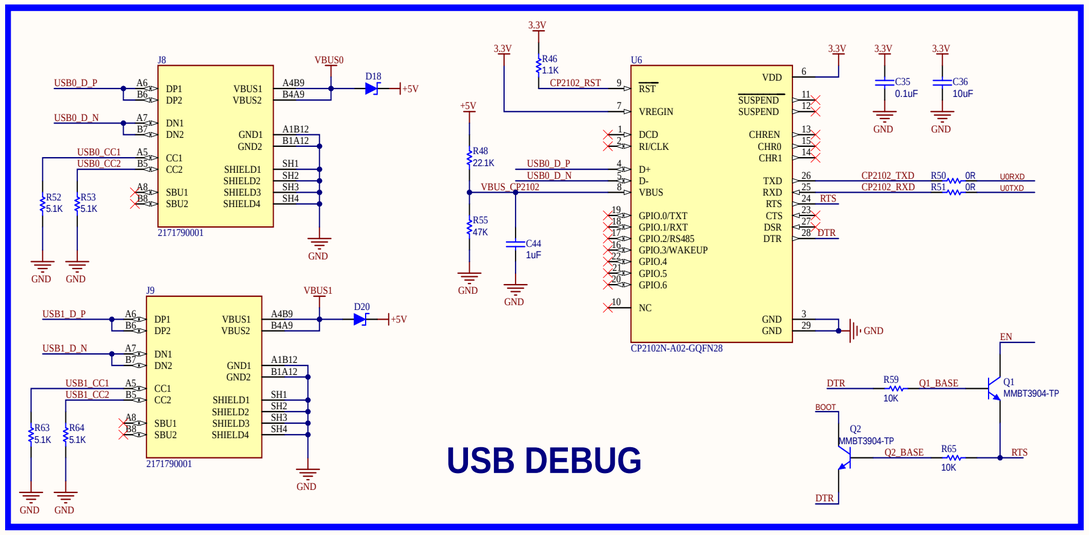

POWER SUPPLY
BAT 6S1P
24V
24V - DC
DC - DC
12V - 2.5A
12V - DC
MOTOR L1
CAN BUS
01
MOTOR L2
CAN BUS
02
MOTOR R1
CAN BUS
03
MOTOR R2
CAN BUS
04
MOTOR L3
UART
MOTOR R3
UART
CAN BUS
CAN IC
CAN IC
DC - DC
3.3V
UART1
UART2
3.3V-DC
DEBUG
USB
USB
UART
AUTO Program
ESP32-S3
I2C
SPI
IMU SENSOR
6500
GPIO
LED
GPIO
BUTTON

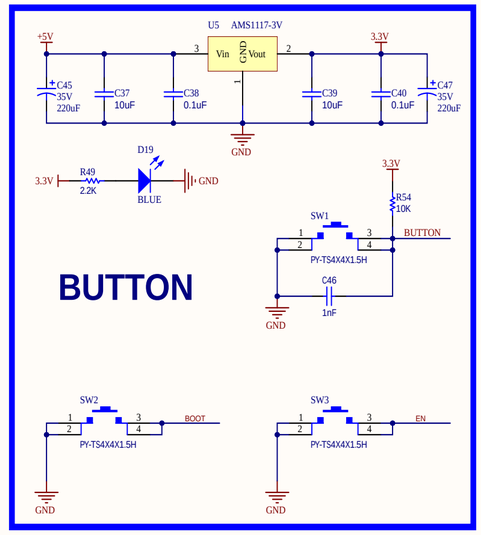

MOTOR
U1
TJA1051T/3/1J
TXD/TXD
GND/GND
VCC
RXD
RS
CANH
CANL
VREF
CAN_TX
CAN_RX
R1 0R
R2 10R
R3 0R
R4 0R
R5 120R
R6 120R
R11 10R
R12 0R
R13 0R
R14 10K
R15 0R
CAN-H
CAN-L
TVS1 SM712-02HTG
C3 0.1uF
C4 10uF
C5 0.1uF
+5V
3.3V
VCC
GND
J2
J3
J4
J5
J6
J7
KF128-5.0_4P
1.0-4P
C9 35V 220uF
C10 35V 220uF
C20 35V 220uF
C21 35V 220uF
C33 35V 220uF
C34 35V 220uF
TVS3 SMBJ22CA-Q
TVS4 SMBJ22CA-Q
TVS5 SMBJ22CA-Q
TVS6 SMBJ22CA-Q
TVS7 SMBJ22CA-Q
TVS8 SMBJ22CA-Q
D5 5V
D6 5V
D7 5V
D8 5V
D9 5V
D10 5V
D11 5V
D12 5V
D14 5V
D15 5V
D16 3V
D17 5V
R41 0R
R43 0R
R42 0R
R44 0R
UART1_RX
UART1_TX
UART2_RX
UART2_TX
24V
12V


POWER
J1
KF128-5.0_4P
D1
D2
D3
C1 35V 220uE
C2 35V 220uF
TVS2 SMBJ22CA-Q
IC1
IC2
TPS54531DDAR
VIN
EN
COMP
RT/CLK
BOOT
SW
FB
GND
C8 0.1uF
C11 2.2uF
C15 35V 220uF
C16 2.2nF
C17 10nF
C22 0.1uF
C23 2.2uF
C27 35V 220uF
C28 35V 220uF
C29 2.2nF
C30 10nF
L1 4.7uH
L2 4.7uH
D4 CMS05(TE12L,Q,M)
D13 CMS05(TE12L,Q,M)
C12 47uF
C13 47uF
C14 47uF
C24 47uF
C25 47uF
C26 47uF
R18 100K
R22 22.1K
R23 37.4K
R31 100K
R36 22.1K
R37 37.4K
R21 100K
R24 18.7K
R34 100K
R38 22.1K
Z1 PTV4.7B-M3_84A
Z2 PTV4.7B-M3_84A
+5V
12V
Iout 2.5A peak 4A


SENSOR
3V3
R7 10K
R8 10K
R9 10K
R10 10K
I2C_SDA
I2C_SCL
SPI_CS
IMU_INT
U2
MPU6050
VCC
GND
SCL
SDA
XDA
XCL
AD0
INT
C6 0.1uF
C7 1uF
R16 0R
R17 0R
R60 0R
R19 0R
R20 0R
ADDRESS I2C : 0x69
U3
MPU6500
FSYC
ECL
EDA
NCS
FSY
C18 0.1uF
C19 1uF
R25 0R
R26 10R
R27 0R
R28 10R
R29 10R
R30 0R
R32 10R
R33 0R
R35 0R
SPI_SCK
SPI_MISO
SPI_MOSI
ADDRESS I2C : 0x68
U4
LSM303DLHC
3.3V
INT2
INT1
DRDY
C31 0.1uF
C32 1uF
R39 0R
R40 0R
R45 0R
ADDRESS I2C : 0x19 & 0x1E


ESP32-S3
U7
ESP32-S3-WROOM-1U-N16R8
R47 10K
C41 10uF
C42 0.1uF
C43 1uF
EN
BUTTON
UART2_TX
UART2_RX
UART1_TX
UART1_RX
I2C_SDA
USB1_D_N
USB1_D_P
U0TXD
U0RXD
LED1
LED2
LED3
R58 2.2K
R62 2.2K
R66 2.2K
D21
D22
D23
BLUE
R56 100K
R57 100K
P1
Header 4X2
BOOT
R61 10K


USB DEBUG
J8
J9
2171790001
USB0_D_P
USB0_D_N
USB0_CC1
USB0_CC2
USB1_D_P
USB1_D_N
USB1_CC1
USB1_CC2
VBUS0
VBUS1
D18
D20
+5V
R52 5.1K
R53 5.1K
R63 5.1K
R64 5.1K
U6
CP2102N-A02-GQFN28
CP2102_RST
VREGIN
VBUS_CP2102
RST
VDD
SUSPEND
DCD
RI/CLK
CHREN
CHR0
CHR1
D+
D-
VBUS
TXD
RXD
RTS
CTS
DSR
DTR
RI
NC
GPIO.0/TXT
GPIO.1/RXT
GPIO.2/RS485
GPIO.3/WAKEUP
GPIO.4
GPIO.5
GPIO.6
R46 1.1K
R48 22.1K
R55 47K
C44 1uF
C35 0.1uF
C36 10uF
CP2102_TXD
CP2102_RXD
U0RXD
U0TXD
R50 0R
R51 0R
R59 10K
R65 10K
Q1 MMBT3904-TP
Q2 MMBT3904-TP
Q1_BASE
Q2_BASE
EN
DTR
RTS
BOOT


BUTTON
U5 AMS1117-3V
Vin
Vout
GND
C45 35V 220uF
C37 10uF
C38 0.1uF
C39 10uF
C40 0.1uF
C47 35V 220uF
D19
BLUE
R49 2.2K
R54 10K
SW1
SW2
SW3
PY-TS4X4X1.5H
C46 1nF
BUTTON
BOOT
EN